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✔ Assignment: Kaggle Competition Peer Review

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Assignment: Kaggle Competition Predict the Sales of the Rossmann Store

November 23, 2015

Shareable Link (<https://www.coursera.org/learn/predictive-analytics/peer/7A5tl/kaggle-competition-peer-review/review/gNlChJG4EeW70BJZtLVfGQ>)

Part 1: Problem Description. Give the name of the competition you selected and write a few sentences describing the competition problem as you interpreted it. You want your writeup to be self-contained so your peer reviewer does not need to go to Kaggle to study the competition description. Clarity is more important than detail. What's the overall goal? What does the data look like? How will the results be evaluated?

Example: "The task is to predict whether a given passenger survived the sinking of the Titanic based on various attributes including age, location of the passenger's cabin on the ship, family members, the fare they paid, and other information. Solutions are evaluated by comparing the percentage of correct answers on a test dataset."



The task is to predict the **Sales** (the output variable representing turnover for any given day) of a given store based on various input variables including Store id, Day of week, Date, the number of customers on a given day, an indicator for whether the store was open, **StateHoliday** (indicating a state holiday), **SchoolHoliday** (indicates if the (Store, Date) was affected by the closure of public schools), a binary variable Promo (indicating whether a store is running a promo on that day) and some other information. Some additional information on store types, assortment levels (basic, extra or extended) along with the sales information. Solutions are evaluated by root mean square percentage error on a test dataset (the lower the better, ideally it should be close to zero), since Sales is a continuous valued variable.

Is the problem description clear and comprehensible?

- ☐ 0 pts
The submission makes no real attempt to describe the problem.
- ☐ 2 pts
The submission makes an attempt to describe the problem, I don't really understand what is being described.
- ☒ 5 pts
The problem description is clear enough; I understand what is going on.



Part 2: Analysis Approach. Write a few sentences describing how you approached the problem. What techniques did you use?

Example: "I split the data by gender and handled each class separately. For the females, I trivially classified all of them as "survived." For the males, I trained a random forest as a classifier. I ignored the pclass attribute that indicated the location of the passenger's cabin because I didn't think it was relevant."

First I pre-processed the dataset. The stores that have the Opened variable set to 0 (are closed) are always going to have 0 sales (since it was closed) on that particular day. So dividing the training dataset into 2 parts, one being the stores that are closed on a particular day (this does not need any model, so it's filtered out) and the other being the ones that are open on a particular day (this data serves as our training dataset and a regression model needs to be trained on this part).

Is the approach to the problem described clearly? Do you have some idea how you might employ these techniques to solve the problem?

- ☐ 0 pts
No real attempt was made to describe the approach to the problem.
- ☐ 2 pts
There's a description, but I don't fully understand what it's saying.
- ☒ 5 pts
The description is clear enough; I understand the approach.



Part 3: Initial Solution. Write a few sentences describing how you implemented your approach. What languages and libraries did you use? What challenges did you run into?

Example: "I partitioned the data by gender manually using Excel. I used Weka to build the random forest."

I partitioned the data by the variable **Open** using **R** and concentrated building regression models for the open stores only. Next, processed the dates to extract out the days months and years from them. Then joined the stores information that were relevant with the train and test dataset and replaced all the NA values in a few variables. I used R to build the elastic net model (generalized linear model with regularization, lasso and ridge regression penalties). I started with this model because of its simplicity (it's linear) and also fast to build. Typically the training dataset was huge (contained 633295 tuples) and to build a non-linear model (such as random forest) typically was very time / resource intensive. I divided the open stores in my training dataset into two parts, to be served as the training and the validation datasets.

Is the initial solution implementation clearly described? Are the tools, languages, and libraries used reasonable?

- ☐ 0 pts
No significant attempt was made to answer the question.
- ☐ 2 pts
The implementation is described, but I don't fully understand how it works.
- ☒ 5 pts
The implementation is well-described; I understand how it works.



Part 4: Initial Solution Analysis. Write a few sentences assessing your approach. Did it work? What do you think the problems were?

Example: "My approach did not work so well, achieving a score of 0.65. This is less than the sample solution. I suspect I should not have ignored the pclass attribute."

My approach did not work so well, achieving a score of 0.60874 (<https://www.kaggle.com/c/rossmann-store-sales/leaderboard?submissionId=2195530>) (<https://www.kaggle.com/c/rossmann-store-sales/leaderboard?submissionId=2195530>). I suspect the linear model is not a good choice for this dataset and we need to use non-linear models. Also, it's better to use some KNN based imputation techniques for the test dataset (since the Open variable contained some NA values), instead of using mean to replace them.

Is the initial solution analysis comprehensible and reasonable?

- ☐ 0 pts
No significant attempt was made to assess the approach.
- ☐ 2 pts
There's an assessment, but I don't really understand what is being described.
- ☒ 5 pts
The assessment is adequately described; I understand what's going on.



Part 5: Revised Solution and Analysis. Write a few sentences describing how you improved on your solution, and whether or not it worked.

Example: "I included the pclass attribute and ignored the ticket number attribute. My score improved to 0.68."

I tried a non-linear model KNN. My score did not improve.

Is the revised solution and analysis presented clearly?

- ☐ 0 pts
No significant attempt was made to describe an improved solution.
- ☒ 2 pts
There's a description, but I don't fully understand it.
- ☐ 5 pts
The improvement is adequately described; I understand it.



Overall evaluation-Given the description provided, do you feel you would be able to reproduce the result?

- ☐ 0 pts
No, the description is too sketchy, incomplete, or unclear for me to be able to reproduce this result.
- ☐ 2 pts
No, but primarily because I don't have a strong enough background in the area; there is a fair amount of jargon used.
- ☒ 4 pts
Yes, I think I could reproduce the result with some effort.
- ☐ 5 pts
Yes, the solution is clearly and thoroughly described -- I could follow them easily.



Evaluating the analyses of others' is a great way to pick up new insights and viewpoints. What is one helpful thing you got out of reviewing this submission? (e.g. maybe ideas on analyzing solutions, incentive to go try this competition, etc.)?



Jeff G. Young

Did you run out of time to describe the revised solution? Otherwise it seemed pretty clear.



Saneil Anup Suri

I did this same competition, so it was interesting to see someone else's approach. I considered creating a separate model for closed stores, so it was interesting to see how that was done.



Christian Thiele

You've picked the right competition to work on, I think the Rossmann competition is currently the most interesting open competition on Kaggle.

(This question will not affect the submitter's score.) You're not expected to actually try and reproduce the submitter's result, but you may choose to do so. Did you try to reproduce the result?

- ☒ No
- ☐ Yes, but my result was different
- ☐ Yes, and my result was the same.



Edit submission

Comments

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